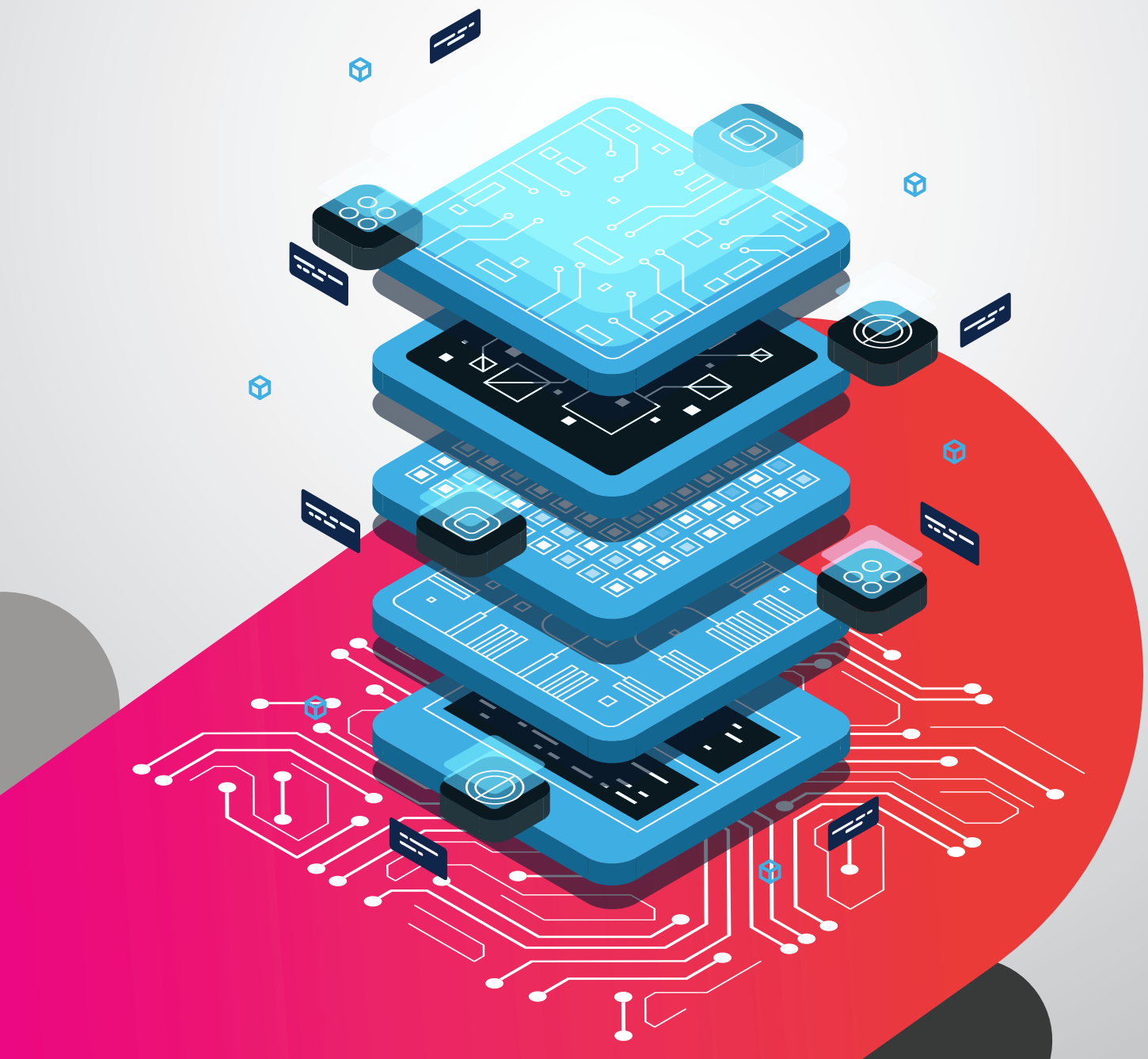
BUSINESSNEXT Technology Stack

Uncover the Engine Behind Innovation

Explore the technologies driving BUSINESSNEXT's intelligent solutions.



Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.6.1 React Framework – 18.3.1 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0
Base OS for Docker Images	Alpine-Based Linux
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token AES-GCM Secret Manager Support with AWS / Azure / GCP
Authentication	LDAP JWT REST API SAML 2.0 Form Authentication API Authentication
Communication with Mail Server	Exchange Online Graph API REST API
API Architecture	BUSINESSNEXT RESTful APIs gRPC

Integrations

System Requirement / Tool	Version / Release
Backend Service Integration	Kafka
Third Party Integration	REST API WEB API SOAP
External DB Connectivity	Supported for Oracle, SQL Server, PostgreSQL, MongoDB, & ClickHouse
Communication Tools	SendGrid ValueFirst Twilio Social Platforms - Meta, LinkedIn, X, YouTube Microsoft Teams & Outlook Integration
Calling Tools	Genesys RingCentral Exotel

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
Web (OS and Web Server)	Linux – Ubuntu (Preferred), Supported for all OS (RedHat for OpenShift)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base Image (Linux OS)	As per Kubernetes provider
HTTPS Certificate for API Security	-
Protocols	Website Communication – HTTP or gRPC File Transfer – FTP
Supported for:	
On Cloud	Azure GCP AWS Oracle Cloud Interface (OCI)
Containerized	OpenShift Tanzu Rancher

Database

System Requirement / Tool	Version / Release
Databases supported for CRMNEXT	
SQL Server	2019 / 2022
Oracle	19c
PostgreSQL	-
Additional Databases Supported	
MongoDB	7.0
Cassandra	-
ClickHouse	-
MinIO	-
Azure Blob Storage	-

Observability (Logging, Tracing, & Monitoring)

System Requirement/Tool	Version/Release
Resource Monitoring (CPU, Memory, Network Bandwidth)	
Grafana	9.3.6
CloudWatch on AWS	-
OpenShift Monitoring Dashboards	-
Google Cloud Monitoring Dashboards (GCP)	-
Other System Requirements	
Elasticsearch	8.3.3
Fluentd	1.16
Kibana	8.3.3
Code Level Metrics	Kibana
Log Time Series Storage Provider	Prometheus
OTEL	As per Kubernetes provider / 0.64.0
Prometheus	2.42.0
JMeter	5.6.3

Development Practices	
Code Commit Approval Process	-
Azure Pipelines being used for Code Comits and approval process	-
Build Pipelines – Automated Triggered overnight)	Automation suite executed Followed by Manual Testing
Stages for Build Pipelines	Compatibility Check Security Check Automation suite executed Followed by Manual Testing Load Testing
Build Release Pipelines	SONAR Scan Image Scan

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.6.1 React Framework – 17.x CSS – 3.0 or later JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 Orléans
Servers	Docker – 20.10.12 Apache Spark 3.5.4 (Used only in DATANEXT)
Web (OS and Web Server)	Linux – RedHat 7.5
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token
Authentication	LDAP JWT REST API SAML 2.0 ADFS API Authentication
API Architecture	BBUSINESSNEXT RESTful APIs gRPC
Backend Services	SendGrid ValueFirst Twilio

Integrations

System Requirement / Tool	Version / Release
RESTful APIs	-
gRPC	-
SendGrid	-
ValueFirst	-
Twilio	-

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Base image Linux OS	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift Tanzu

Database

No additional database is required. Support is provided for the following:

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
PostgreSQL	-
Oracle	19c

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
OTEL	As per Kubernetes provider
EFK - App / Hardware	Kibana (Latest Version)

Advanced Analytics

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.6.1 React Framework – 16.13.1 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
CRMNEXT Database	Mandatory (required even when using ClickHouse)
Kafka	-
Internal Consumer Services	-
ClickHouse	25.8

Integrations

System Requirement / Tool	Version / Release
CDC Consumer Service (for dumping data into ClickHouse)	-

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7
PostgreSQL	-
ClickHouse	-

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
OTEL	As per Kubernetes provider

Frontend

System Requirement / Tool	Version / Release
Programming Language	jQuery – 3.6.1 React Framework – 16.13.1 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 Orléans
Servers	Docker – 20.10.12 Apache Spark 3.5.4 (Used only in DATANEXT)
Web (OS and Web Server)	Linux – RedHat 7.5
Security	AE-256 RSA
Authentication	LDAP
API Architecture	BUSINESSNEXT RESTful APIs gRPC

Integrations

System Requirement / Tool	Version / Release
ValueFirst	-
Twilio	-

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
OTEL	As per Kubernetes provider

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.7.1 React Framework – 18.2.0 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12 Python 3.11
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 Trino Hive Apache Iceberg Apache Spark 3.5.4
Security	TLS 1.2 RSA SHA-2 AES-256
API Architecture	RESTful APIs gRPC

Integrations

System Requirement / Tool	Version / Release
RESTful APIs	11.*.*
gRPC	1.27

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift Rancher

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-
ClickHouse	25.8.1
Cassandra	-
MinIO	RELEASE.2025-07-23T15-54-02Z
AWS S3	-
NetApp StorageGRID	-

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log Time-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.6.1 React Framework – 16.13.1 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12 Python 3.11
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 scikit-learn Dask ONNX Apache Spark 3.5.4 (Used only in DATANEXT)
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token
Authentication	LDAP JWT REST API SAML 2.0 ADFS
API Architecture	RESTful APIs gRPC

Integrations

System Requirement / Tool	Version / Release
RESTful APIs	11.*.*
gRPC	1.27

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	Open Shift

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-
AWS S3	-
Milvus (Vector DB)	2.5.13
MinIO	RELEASE.2025-07-23T15-54-02Z

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log Time-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.7.1 React Framework – 18.2.0 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12, Python 3.11
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 LangChain Transformers LLM Apache Spark 3.5.4 (Used only in DATANEXT)
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token
Authentication	LDAP JWT REST API SAML 2.0 ADFS
API Architecture	RESTful APIs gRPC

Integrations

System Requirement / Tool	Version / Release
RESTful APIs	11.*.*
gRPC	1.27

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-
Azure Blob Storage	-
Milvus (Vector DB)	2.5.13
MinIO	RELEASE.2025-07-23T15-54-02Z

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log Time-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.7.1 React Framework – 18.2.0 CSS – 3.0 JavaScript – ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12 Python 3.11
Framework	.NET 8 (Feb 2025, build version 11.31) MVC 6.0 ADO.NET Orléans LLM LangGraph
Backend Service Integration	Kafka
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token
Authentication	LDAP JWT REST API SAML 2.0 ADFS

Integrations

System Requirement / Tool	Version / Release
RESTful APIs	11.*.*
gRPC	1.27

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-
Milvus (Vector DB)	2.5.13
AWS S3	-
MinIO	RELEASE.2025-07-23T15-54-02Z

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log Time-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider

CUSTOMERNEXT

BUSINESS NEXT

Frontend

System Requirement / Tool	Version / Release
Programming Language	HTML – 5.0 jQuery – 3.7.1 React Framework – 18.2.0 CSS – 3.0 JavaScript - ES6

Backend

System Requirement / Tool	Version / Release
Programming Language	C# (C-sharp) – 12
Framework	.NET 8 (Feb 2025, build version 11.31) ADO.NET MVC 6.0 Next.js Apache Spark 3.5.4 (Used only in DATANEXT)
Security	TLS 1.2 RSA SHA-2 AES-256 Anti-CSRF Token
Authentication	LDAP JWT REST API

Integrations

System Requirement / Tool	Version / Release
RESTful APIss	11.*.*
gRPC	1.27
Kafka	-

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift Tanzu

Database

System Requirement / Tool	Version / Release
SQL Server	2019 / 2022
Oracle	19c
MongoDB	7.0
PostgreSQL	-
Azure Blob Storage	-
Milvus (Vector DB)	-

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log Time-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider
Control Surface	-

Note:

The information provided pertains exclusively to custom AI-based APIs and custom UI. No specific details have been included regarding BUSINESSNEXT, the platform on which the Lending workflow is configured.

Frontend

System Requirement / Tool	Version / Release
Framework	.NET Core - 6.0.36 Angular - 20.2.4 React - 19.1.1 Next.js - 15.5.2 TypeScript - 5.9.2
Development Tool	VS Code (Latest)

Backend

System Requirement / Tool	Version / Release
Node.js	22.17.1
Next.js	15.5.2
API Architecture	REST API
Security	TLS 1.2 RSA SHA-2 AES-256 OAuth 2.0 JWT Authentication

API

System Requirement / Tool	Version / Release
Programming Language - Python	3.12
Pytesseract	0.3.13
TensorFlow	2.20.0
OpenCV	4.12.0.88
Ultralytics	8.3.193
pyodbc	5.2.0

Integrations

System Requirement / Tool	Version / Release
LENDINGNEXT Platform	-
RESTful APIs	11.*.*
gRPC	1.27

Deployment

System Requirement / Tool	Version / Release
Kubernetes Distribution	1.15.5
OS for Kubernetes	Linux RedHat 7.5 or Ubuntu 22.04 LTS
Docker	19.3.1
Kubernetes Persistent Volume Provider	Rook / Ceph (Latest Version)
HTTP Proxy (NGINX)	As per Kubernetes provider
Base image (Linux OS)	As per Kubernetes provider
HTTPS certificate for API security	-
Supported for:	
On Cloud	Azure GCP AWS
Containerized	OpenShift Tanzu

Database

Following databases are supported:

System Requirement / Tool	Version / Release
SQL Server	2019/2022
Oracle	19c
PostgreSQL	-

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Log TTime-Series Storage Provider	Prometheus
EFK - App / Hardware	Grafana / Kibana (Latest Version)
OTEL	As per Kubernetes provider
Control Surface *	-

*Control Surface is applicable only in scenarios where CUSTOMERNEXT is implemented for lending workflows.

MOBILE LITE

Frontend

System Requirement / Tool	Version / Release
Programming Language	Dart - 3.7.2
UI Toolkit	Flutter - 3.29.3
Android Development	Kotlin - 2.0.0 Java SE - 17.0.11
iOS Development	Swift - 5.0
Database	Floor Drift
Android SDK	minSdk = 31 targetSdk = 35
iOS SDK	mindeployment = 17.0

Backend

System Requirement / Tool	Version / Release
The mobile application consumes REST API services developed in .NET and forms an integral part of the CRMNEXT core.	The version is tightly coupled with the specific Mobile and CRMNEXT build.

Integrations

System Requirement / Tool	Version / Release
Firebase Support: • Google Map / OLA / MapMyIndia • Google Analytics • Firebase Crashlytics	Latest

Deployment

System Requirement / Tool	Version / Release
Play Store / App Store	Latest
MDM	Latest

Database

System Requirement / Tool	Version / Release
Drift (Internal Package)	2.26.1

Observability (Logging, Tracing & Monitoring)

Requirement	Tool / Framework
Firebase Crashlytics	Latest